UNIVERSITY OF MAKATI
COLLEGE OF COMPUTING AND INFORMATION SCIENCES
Bachelor of Science in Information Technology

HULOG: IOT-ENABLED DYNAMIC QR PARCEL DROP VAULT FOR RESIDENTIAL GATES AND BOARDING HOUSES

A Project Research Status Report Submitted to the Faculty of the
College of Computing and Information Sciences
University of Makati

In Partial Fulfillment of the Requirements for the Course
Technopreneurship & Real-Time Systems (RTS)
Academic Year 2026-2027

Course Professor:
Engr. Bhai Nhuraisha I. Deplomo, PECE, MSEcE, MSCIT

Researchers / Proponents:
Camarador, Jhorenz S.
Corpuz, Lance Josh M.
Goze, AJ M.
Luzano, Justin Mark D.

16 October 2026

EXECUTIVE SUMMARY

Daily courier stops have become part of household life in Philippine cities, and the fragile moment is the last one: the rider at a gate when nobody is home. Parcels are left on a fence ledge, dropped over the gate, or passed to a neighbor. Rain, heat, and theft sit on the other side of that habit. A padlock cannot be lent to a different rider every afternoon. A QR sticker on a bin can be photographed once and reused later.

HULOG is a drop vault for a house gate or a boarding-house entrance. The owner creates a 6-digit PIN and a matching QR pass from a web page. The pass is designed to live for 15 or 30 minutes and to die after one completed door cycle. The rider types the PIN on a keypad, or opens the pass in the phone browser, and does not install an app. A 12 V door solenoid, switched by a relay, releases the latch. A magnetic reed switch and an infrared proximity sensor are there to confirm that the door opened, that a parcel is in the bay, and that the door shut again.

As of 16 October 2026 the purchased parts are in hand, the web portal and the hardware simulator run, and the ESP32 sketch on the bench already reads the keypad, the reed switch, and the proximity sensor, drives the OLED, and pulses the relay. That sketch still accepts a fixed test PIN. Wi-Fi, the expiring pass, and the Telegram notice are specified and are not in the flashed program yet. Six completed checkouts for the lock, sensors, power, display, and wire came to PHP 1,185.00. The listed product prices inside those orders sum to PHP 1,124.00. The ESP32 development board, the GPIO expansion board, and the 4x4 membrane keypad are parts to buy at PHP 350.00, PHP 119.00, and PHP 50.00. The listed product prices for all fourteen parts come to PHP 1,643.00.

CHAPTER 1: INTRODUCTION AND PROBLEM CONTEXT

1.1 Problem Background

The Department of Trade and Industry, in the logistics section of the E-Commerce Philippines 2022 Roadmap, treats last-mile delivery as a constraint on online trade. The roadmap names a shortage of motorcycle couriers, worsening traffic, and the extra risk riders carry when they must collect cash on delivery (Department of Trade and Industry, 2022). Those are system problems. The failure this project is built for is smaller and closer to the door.

In a gated house, a townhouse, or a student boarding house, the recipient is often at work or in class during the hours when Shopee Xpress, J&T Express, Flash Express, Grab, and Lalamove riders are on the street. The rider is on a quota and will not wait. The usual fixes fail for mechanical reasons. A key cannot be passed safely to a rotating set of riders. A static PIN, once seen, stays valid until someone remembers to change it. A printed QR code is only an image: whoever photographs it can present it again.

What the household needs is a box at the gate that a stranger can open once, for a short time, without an account in a courier app, and that can show the owner the door was shut on a parcel. The box also has to be buildable from parts sold in Metro Manila.

1.2 Objectives

The project will design, build, and evaluate HULOG, a household parcel vault that accepts one delivery through a short-lived PIN or QR pass while the recipient is away.

1. **Lock drive.** Switch a 12 V door solenoid from an ESP32 through a 1-channel optocoupler relay, so the microcontroller pin never carries the lock current.
2. **Door and parcel check.** Use the normally open lead of a magnetic reed set, and an infrared proximity module aimed at the bay, before a pass is treated as spent.
3. **Single-use pass.** Issue a 6-digit PIN and a QR symbol with a life of 15 or 30 minutes, and invalidate that pass after one completed door cycle. This is the design target. The bench sketch does not enforce it yet.
4. **Two web pages.** Give the owner a page that mints passes, and give the rider a browser page that needs no installed app.
5. **One adapter, two rails, real prices.** Feed the lock from a 12 V 2 A adapter and the logic from an LM2596 set to 5 V, and report the cost from the receipts rather than from a catalog guess.

1.3 Significance of the Study

- **Households and boarding-house tenants** get a place to receive a prepaid parcel when they are not at the gate.
- **Riders and sellers** get a stop that can end without a wait, plus a door-and-sensor record that a porch photo does not provide.
- **Local builders** get a worked example of a gate box made from modules sold in Manila, with the peso amounts taken from completed orders.

1.4 Scope and Delimitation

The prototype is a tabletop vault for a small or medium prepaid parcel. The controller is an ESP32 development board seated on a GPIO expansion board. The expansion board does not create new GPIO numbers. It brings the existing pins out to headers that are easier to wire, which is why the kit includes both female-to-female and female-to-male jumpers. The door carries a 4x4 membrane keypad and a 0.96-inch SSD1306 OLED (128x64, blue). The owner's page and the rider's page are a React, Vite, and TypeScript application. Telegram is the planned notice channel.

The vault does not cool food. Cash-on-delivery is outside the cycle, because the box cannot take money. There is no GSM radio. The intended link is household Wi-Fi. The program on the bench today does not join that network yet. The wooden or acrylic enclosure is not on the receipts and is scheduled after this report.

CHAPTER 2: REVIEW OF RELATED LITERATURE AND STUDIES

2.1 The Last Mile, and the Philippine Constraint

Gevaers, Van de Voorde, and Vanelslander (2014) describe the last mile of business-to-consumer delivery as one of the more expensive, least efficient, and most polluting stretches of the whole chain. Their point is about cost and wasted trips, not about a specific city. It still names the stretch this project sits on: the final handoff, where a failed meeting between rider and recipient turns into another attempt.

The Philippine policy document that speaks directly to that stretch is the Department of Trade and Industry's E-Commerce Philippines 2022 Roadmap. Its logistics page says e-fulfillment and last-mile service are held back by a lack of motorcycle couriers, by traffic congestion, and by the risk cash-on-delivery places on the rider (Department of Trade and Industry, 2022). The roadmap does not publish a count of parcels left on gates. That local picture, in Section 1.1, is the setting the proponents are designing for. The roadmap is used here only for the constraints it actually states.

2.2 Lockers, Pickup Points, and a Box at the House

Morganti, Seidel, Blanquart, Dablanc, and Lenz (2014) compare two European answers to home delivery: a pickup counter inside a shop, and an automated locker bank. They note that shoppers still prefer delivery to the door, and that this preference breaks parcel flow into many small urban stops. Iwan, Kijewska, and Lemke (2016) study InPost lockers in Poland and treat the locker as a way to make the courier's round more productive. In both studies the household still has to travel to a shared machine. A locker on a petrol-station forecourt does not solve a boarding house whose residents are in class, and a dense barangay often has no sidewalk wide enough for a bank of steel doors.

Punakivi, Yrjola, and Holmstrom (2001) move the box to the house. They compare a reception box installed at the home with a carrier-owned delivery box. In their grocery simulation, unattended reception cut home-delivery transport cost by up to 60 percent against attended delivery. They also show the limit of that result. The saving depended on the resident paying for and keeping a box, and the reception box they modeled was often refrigerated. HULOG keeps the idea that the secure point is at the home, and drops the cold box. A shirt or a phone case does not need refrigeration, and Section 1.4 excludes groceries for that reason. What the 2001 study does not give is a way for a rotating set of local riders to open the door without a key and without a dedicated courier application.

2.3 A Code That Must Expire

Focardi, Luccio, and Wahsheh (2019) show that an ordinary QR symbol has no standard proof of who created it or whether it is still the symbol the owner issued. People scan what is in front of them. For a parcel box, that finding has a direct physical form: a sticker that never changes can be copied with a camera and shown again on another day. A fixed keypad PIN has the same shape of failure. Once a rider has seen it, it remains valid until a person changes it.

M'Raihi, Machani, Pei, and Rydell (2011), in RFC 6238, specify the time-based one-time password. The code is valid only inside a short time step, so a value someone has already seen does not stay useful. HULOG's design follows that limit and adds a stricter one. The pass should also be marked used when the door cycle finishes, so it cannot be replayed inside the time window. The bench sketch does not compute RFC 6238. It compares the keys with a fixed test PIN. Section 3.7 states that gap in the open, because a status report that hides it would describe a different machine from the one on the bench.

2.4 Why the Lock Is Not on a Timer Alone

A solenoid that releases for a few seconds and then drives again, with no knowledge of the door, can extend the bolt while the door is still open. The next slam seats the bolt against the frame instead of into the strike. A notice sent at the moment the PIN is accepted can also report a delivery that never entered the box. The reed switch answers a yes-or-no question about the door: it left the frame, and it returned. The proximity module answers a different question: an object is in front of the sensor, in the bay. The purchased module is a reflective infrared sensor, not a separate lamp and detector across the opening. It has to be aimed at the spot where the parcel sits. The pass should be spent only after both answers have arrived, in that order. That rule is the design. The bench sketch already reads both sensors. It does not yet retire a minted pass, because it does not yet hold one.

2.5 Synthesis and Research Gap

The literature supports three separate claims, and it does not assemble them into one device. The last mile is costly and, in the Philippines, operationally tight (Gevaers et al., 2014; Department of Trade and Industry, 2022). Shared lockers help the courier and still ask the resident to travel (Morganti et al., 2014; Iwan et al., 2016). A box at the house can remove that trip, at the resident's expense, and the studied version was aimed at cold groceries (Punakivi et al., 2001). A visible QR code is easy to copy because the symbol itself carries no proof of origin (Focardi et al., 2019), and a time-step password is the standard remedy for a code that must die (M'Raihi et al., 2011).

The gap HULOG is built to fill is a non-refrigerated box at a Philippine gate or boarding-house door, opened by a pass the rider already has in a message, with no courier application, using the modules in Table 1. The cost claim in this report is the listed product prices in Table 1, PHP 1,643.00. Selling price, labor, and the enclosure belong in Chapter 4, after those costs exist.

CHAPTER 3: METHODOLOGY AND TECHNICAL DESIGN

3.1 Parts to Buy

Table 1 is the parts list to buy. Sellers are MakerLab, Tool Corner, and ENG LAB.ph. Items 1-11 are the product prices on the completed orders. Items 12-14 are the ESP32 development board, the GPIO expansion board, and the 4x4 membrane keypad, at the MakerLab price. The listed product prices for items 1-14 come to PHP 1,643.00.

Table 1

Parts to buy

No.	Component	Seller	Qty	Unit (PHP)	Line (PHP)
1	Active buzzer driver module High-current module, blue board	MakerLab	1	35.00	35.00
2	LM2596 buck converter DC-DC step-down module	MakerLab	1	49.00	49.00
3	Dupont jumpers, female-to-female 20 cm set	MakerLab	1	59.00	59.00
4	Dupont jumpers, female-to-male 20 cm set	MakerLab	2	59.00	118.00
5	Solenoid door lock 12 V DC, for doors	MakerLab	1	310.00	310.00
6	Relay module 1-channel, 5 V coil, optocoupler	MakerLab	1	49.00	49.00
7	DC power adapter 12 V, 2 A	MakerLab	1	145.00	145.00
8	Infrared proximity module Reflective obstacle sensor	MakerLab	1	28.00	28.00
9	DC barrel jack, female 5.5 x 2.1 mm, screw terminals	Tool Corner	3	7.00	21.00
10	OLED display 0.96 in, SSD1306, 128x64, blue, I2C	ENG LAB.ph	1	145.00	145.00
11	Magnetic door reed set NO and NC leads, with magnet	MakerLab	1	165.00	165.00
12	ESP32 development board 30-pin or 38-pin Wi-Fi board	MakerLab	1	350.00	350.00
13	GPIO expansion board Terminal board for the 30-pin or 38-pin ESP32. Brings the existing pins out to headers.	MakerLab	1	119.00	119.00
14	Membrane keypad 4x4, 16 keys	MakerLab	1	50.00	50.00
	Listed product prices, items 1-14				**1,643.00**

The listed product prices for items 1-14 come to PHP 1,643.00. Six completed checkouts for items 1-11 were PHP 583.00 (buzzer, buck, both jumper sets, and solenoid), PHP 202.00 (relay and adapter), PHP 28.00 (proximity module), PHP 53.00 (three jacks), PHP 145.00 (OLED), and PHP 174.00 (reed set). Those payments sum to PHP 1,185.00. The PHP 61.00 difference from the PHP 1,124.00 product prices on those orders is shipping and other checkout charges, mostly on the jack order, whose price was PHP 7.00 each after a discount from PHP 25.00. Items 12-14, the ESP32, the GPIO expansion board, and the keypad, are included in the PHP 1,643.00 at the MakerLab prices in the table. The enclosure, hinges, and the 1N4007 flyback diode are not in this list. The diode is specified in Figure 2 and should be fitted before the coil is pulsed repeatedly.

3.2 How a Drop Is Arranged

Figure 1 is the arrangement, not a claim that every arrow is already live. The owner's page and the rider's page exist. The link from the vault to Telegram does not. The ESP32 drawn at the side of the box is the board on the expansion headers. The keypad and the OLED are on the outside of the door, where the rider can reach them. The reed set sits on the door gap. The proximity module looks into the bay. The solenoid sits at the strike.

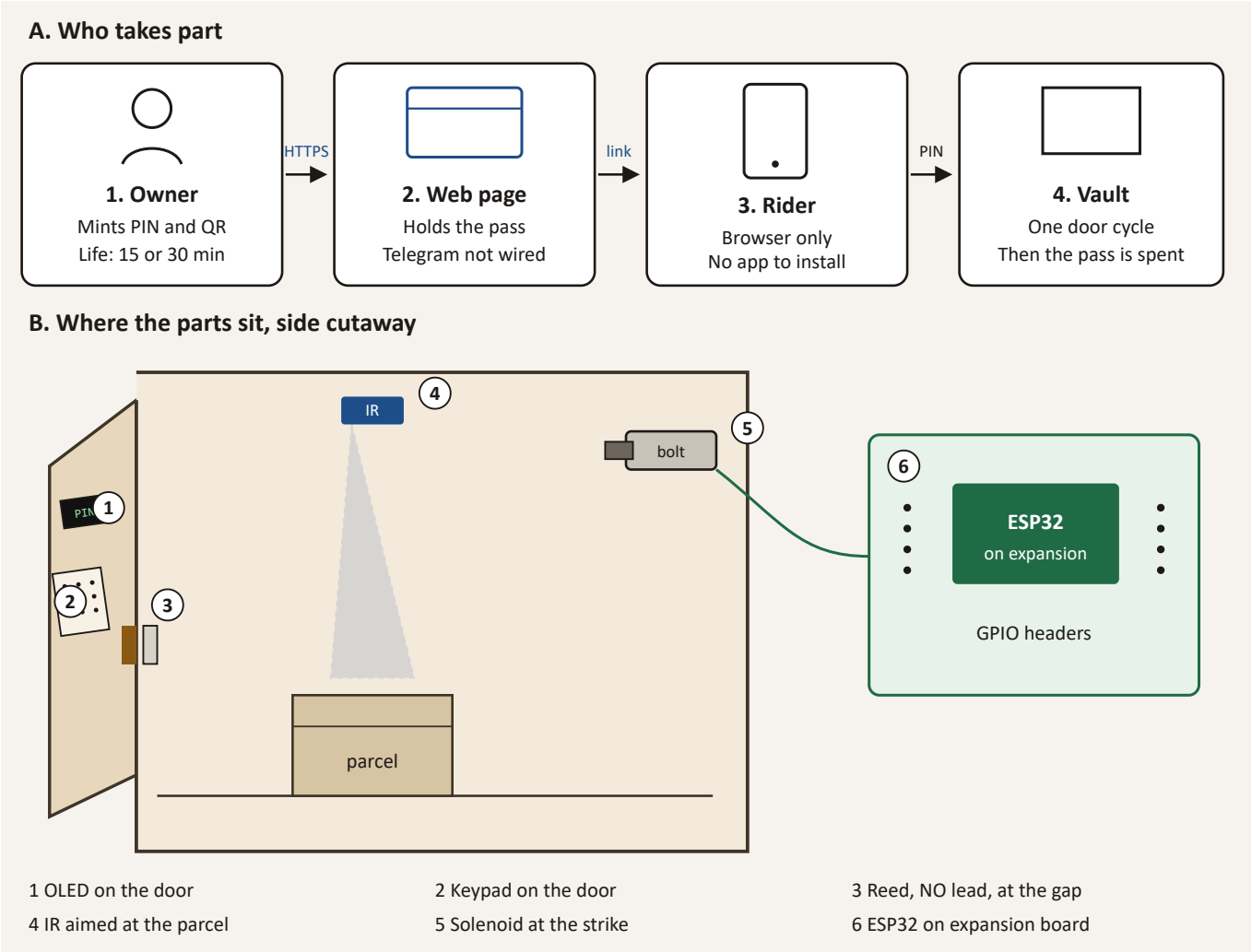


Figure 1. People, pass, and part placement for one drop. The Telegram link in step 2 is drawn as a plan. It is not in the flashed sketch.

3.3 Power and Lock Circuit

One 12 V 2 A adapter feeds two rails, drawn in Figure 2. The red rail is the lock. It goes to the relay's common contact, through the normally open contact, and through the solenoid coil to ground. The blue rail is logic. An LM2596 module steps 12 V down to 5 V for the ESP32 VIN pin, the relay-module supply, and the buzzer module. The OLED and the proximity module take 3.3 V from the ESP32, so their outputs cannot rise above the pin rating. GPIO 23 goes only to the relay input. It is an active-low signal. It does not join the red wire.

The LM2596 often ships with its output set near the input voltage. The output is measured and set to 5.0 V before the ESP32 is connected. The solenoid's coil current is not printed on the receipt. It will be measured on the bench before long pulses, against the 2 A rating of the adapter. A 1N4007 is drawn across the coil, band toward the 12 V side, so the spike from the collapsing field has a path that is not the relay contact. That diode was not in the orders.

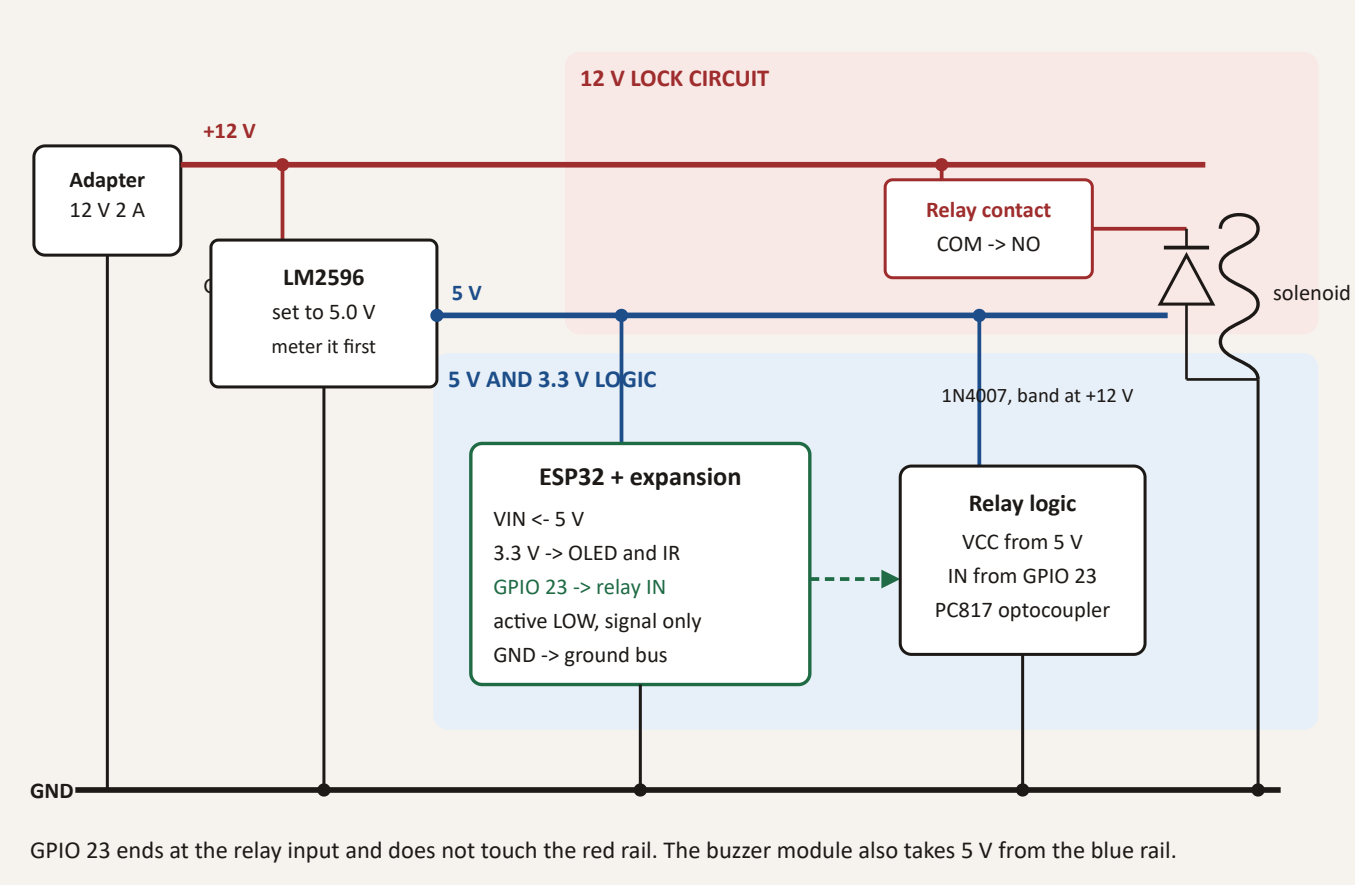


Figure 2. Dual-rail supply and lock circuit. The optocoupler keeps the ESP32 pin on the logic side of the relay. The flyback diode is specified and was not on the receipts.

3.4 Pin Map

Table 2 is the map in the bench sketch. GPIO 0, GPIO 2, and GPIO 15 are left free because they are sampled at boot. GPIO 12 is also a boot pin, and the keypad does use it as row 2. That row must sit low while the board starts. If it is pulled high at reset, the ESP32 can fail to boot. The reed set sold to the group has both a normally open lead and a normally closed lead. The sketch uses the normally open lead with the internal pull-up, so GPIO 19 reads low when the magnet is against the switch, which is the shut door. The normally closed lead is left unconnected.

Table 2

ESP32 GPIO map used by the bench sketch

Device	ESP32 pin	What the wire does
Relay input	GPIO 23	Output, active low. Signal into the optocoupler only.
OLED SDA / SCL	GPIO 21 / GPIO 22	I2C. Power the display from the 3.3 V pin.
Reed switch, NO lead	GPIO 19	Input with pull-up. Low means the door is shut.
IR proximity output	GPIO 18	Input. Low means an object is in front of the sensor. Power the module from 3.3 V.
Buzzer module input	GPIO 5	Output. High sounds the module. Module supply is the 5 V rail.
Keypad rows	GPIO 13, 12, 14, 27	Row scan. GPIO 12 must not be high at boot.
Keypad columns	GPIO 26, 25, 33, 32	Column inputs with internal pull-ups.
LM2596 output	VIN and GND	5.0 V into the board's 5 V input, after the meter check. Not into a GPIO pin.

If the proximity module will not detect at 3.3 V on the bench, it may be powered from 5 V only with a resistor divider on its output before GPIO 18. A 5 V high straight into an ESP32 pin is outside the pin rating.

3.5 What the Bench Sketch Actually Does

Figure 3 follows *firmware/src/main.cpp* as it stands. Solid boxes run today. The dashed box is the design target from Section 1.2 and is not in the flashed loop. A 30-second timeout returns the vault to idle if the door cycle is not finished. If the door shuts and the infrared sensor never saw a parcel, the sketch returns to idle and does not treat the cycle as a delivery. The confirmation screen says the door cycle is done. It does not say that a Telegram message was sent.

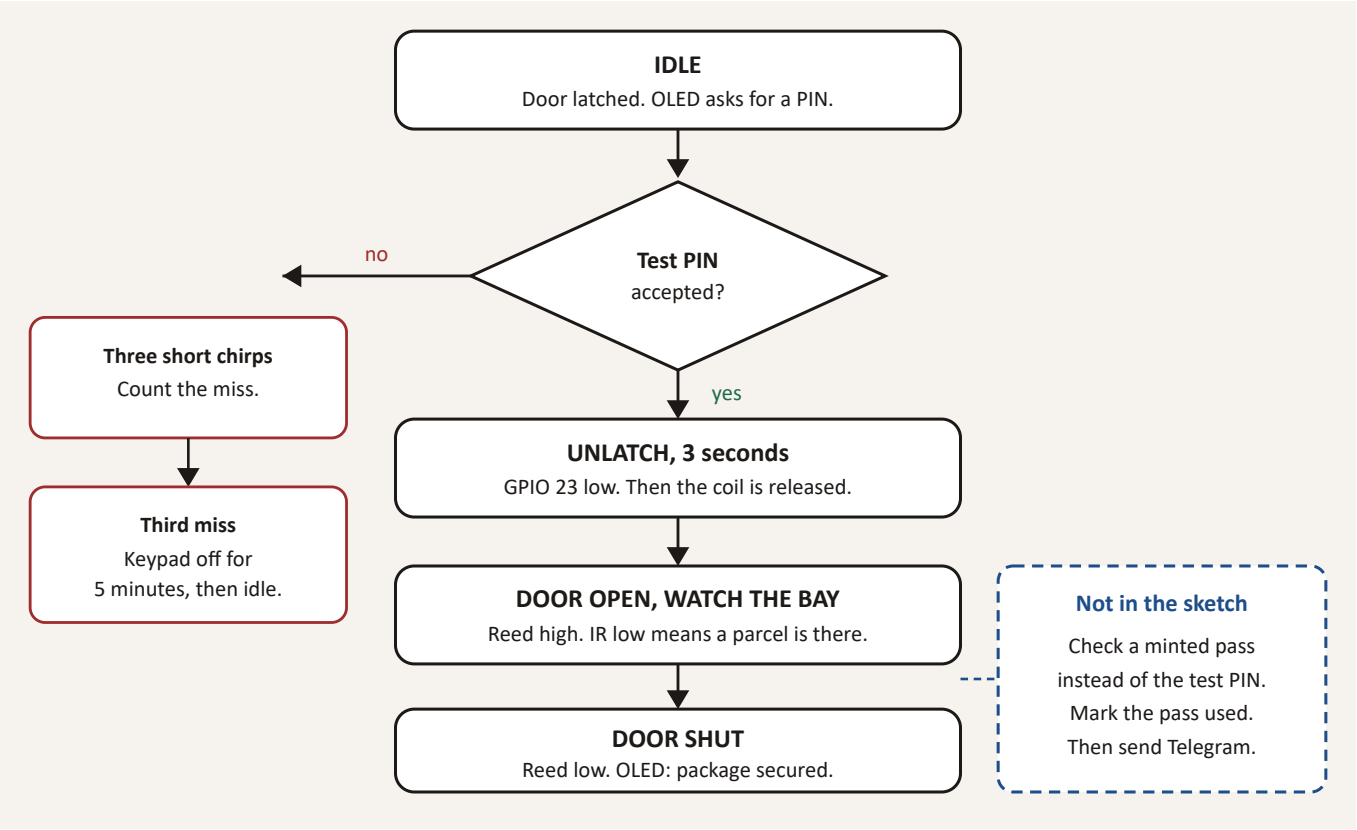


Figure 3. Door cycle in the current firmware. Expiring passes and Telegram are specified beside it and are not executed by this loop.

3.6 Software Beside the Sketch

The web application in *hulog-web/* is the owner's page, the rider's browser page, and a simulator of the vault. It is built with React 18, Vite, and TypeScript. The design target for a pass is 15 or 30 minutes. The simulator still uses a longer demonstration lifetime, so a defense demo of the page is not yet proof that the 15-minute rule is enforced. The firmware is an Arduino-framework sketch under PlatformIO. It uses the SSD1306 library for the display and the keypad library for the matrix. It does not open Wi-Fi, it does not write a pass to flash, and it does not call Telegram.

3.7 Implementation Status

Table 3 is the status on 16 October 2026. A row marked designed is a specification, not a finished test.

Table 3

What is done, what is only specified

Piece	Standing	Evidence
Receipt parts, items 1-11	Purchased	Six completed orders, PHP 1,185.00 paid.
ESP32, expansion board, keypad	To buy	Listed in Table 1 at PHP 350.00, PHP 119.00, and PHP 50.00. The bench sketch is written for this board, expansion, and keypad.
Repository and notes	In use	Public repository *buridekz/hulog-iot-vault*, with wiring and architecture notes.
Web portal and simulator	Built	*hulog-web/*. The simulator lifetime is still a demo window.
Bench sketch	Running its loop	Keypad, OLED, reed, IR, 3-second relay pulse, 5-minute lockout, fixed test PIN.
Wi-Fi, minted pass, Telegram	Designed	No network calls in the flashed sketch. Figure 3 marks this box as dashed.
Flyback diode	Specified	Not on the receipts. Fit a 1N4007 before repeated coil pulses.
Enclosure	Not started	No box or hinges on the receipts. Scheduled after this report.

3.8 Next Steps for Part 2

1. Set the LM2596 to 5.0 V with a meter, then wire the expansion board, relay, reed (NO lead), proximity module, OLED, buzzer, and solenoid as in Figure 2 and Table 2.
2. Fit the 1N4007 across the coil. Measure the coil current. Confirm that GPIO 12 is low at boot.
3. Run the door cycle on the bench: good PIN, bad PIN three times, door shut with no parcel, door shut with the IR covered, and the 30-second timeout.
4. Replace the fixed test PIN with the minted 15- or 30-minute pass, then send the Telegram notice only after the door-and-parcel check.
5. Build the tabletop enclosure and write Chapter 4 from the PHP 1,643.00 listed in Table 1, plus the enclosure and labor. Do not price the product from the electronics lines alone.

REFERENCES

Department of Trade and Industry. (2022). *E-Commerce Philippines 2022 roadmap: Logistics*. https://ecommerce.dti.gov.ph/madali/logistics.html

Focardi, R., Luccio, F. L., & Wahsheh, H. A. M. (2019). Usable security for QR code. *Journal of Information Security and Applications, 48*, Article 102369. https://doi.org/10.1016/j.jisa.2019.102369

Gevaers, R., Van de Voorde, E., & Vanelslander, T. (2014). Cost modelling and simulation of last-mile characteristics in an innovative B2C supply chain environment with implications on urban areas and cities. *Procedia - Social and Behavioral Sciences, 125*, 398-411. https://doi.org/10.1016/j.sbspro.2014.01.1483

Iwan, S., Kijewska, K., & Lemke, J. (2016). Analysis of parcel lockers' efficiency as the last mile delivery solution - The results of the research in Poland. *Transportation Research Procedia, 12*, 644-655. https://doi.org/10.1016/j.trpro.2016.02.018

M'Raihi, D., Machani, S., Pei, M., & Rydell, J. (2011). *TOTP: Time-based one-time password algorithm* (RFC 6238). Internet Engineering Task Force. https://doi.org/10.17487/RFC6238

Morganti, E., Seidel, S., Blanquart, C., Dablanc, L., & Lenz, B. (2014). The impact of e-commerce on final deliveries: Alternative parcel delivery services in France and Germany. *Transportation Research Procedia, 4*, 178-190. https://doi.org/10.1016/j.trpro.2014.11.014

Punakivi, M., Yrjola, H., & Holmstrom, J. (2001). Solving the last mile issue: Reception box or delivery box? *International Journal of Physical Distribution & Logistics Management, 31*(6), 427-439. https://doi.org/10.1108/09600030110399423